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Chapter 1

Basics

This chapter covers fundamental concepts that should already be familiar from high school studies.

Basic Integration Formulae

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C, n \neq -1$$

$$\int dx = x + C$$

$$\int \cos x dx = \sin x + C$$

$$\int \sin x dx = -\cos x + C$$

$$\int \sec^2 x dx = \tan x + C$$

$$\int \csc^2 x dx = -\cot x + C$$

$$\int \sec x \tan x dx = \sec x + C$$

$$\int \csc x \cot x dx = -\csc x + C$$

$$\int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x + C$$

$$\int \frac{dx}{\sqrt{1-x^2}} = -\cos^{-1} x + C$$

$$\int \frac{dx}{1+x^2} = \tan^{-1} x + C$$

$$\int \frac{dx}{1+x^2} = -\cot^{-1} x + C$$

$$\int \frac{dx}{x\sqrt{x^2-1}} = \sec^{-1} x + C$$

$$\int \frac{dx}{x\sqrt{x^2-1}} = -\csc^{-1} x + C$$

$$\int e^x dx = e^x + C$$

$$\int \frac{dx}{x} = \log |x| + C$$

$$\int a^x dx = \frac{a^x}{\ln a} + C$$

Ex. 1

$$\int x^2 (x^{1/2} + x^{-1/2} + x^{-3/2}) dx = \int (x^{5/2} + x^{3/2} + x^{1/2}) dx = \frac{2}{7}x^{7/2} + \frac{2}{5}x^{5/2} + \frac{2}{3}x^{3/2} + C$$

Ex. 2

$$\int \frac{x^4 + 2x^2 + 2}{x^2 + 1} dx = \int \left(x^2 + 1 + \frac{1}{x^2 + 1} \right) dx = \frac{x^3}{3} + x + \arctan x + C$$

Ex. 3

$$\int \frac{(1+x)^2}{x^2} dx = \int \frac{x^2 + 2x + 1}{x^2} dx = \int \left(1 + \frac{2}{x} + \frac{1}{x^2} \right) dx = x + 2 \ln |x| - \frac{1}{x} + C$$

Ex. 4

$$\int 2^x (1 + 5^x) dx = \int (2^x + 10^x) dx = \frac{2^x}{\ln 2} + \frac{10^x}{\ln 10} + C$$

Ex. 5

$$\int (e^x + \sin x) dx = e^x - \cos x + C$$

Ex. 6

$$\int (\csc^2 x + \csc x \cot x) dx = -\cot x - \csc x + C$$

Fundamental Theorem of Calculus

Part I If f is continuous on $[a, b]$, then the function $F(x) = \int_a^x f(t) dt$ is continuous on $[a, b]$, differentiable on (a, b) , and $F'(x) = f(x)$.

Part II If f is continuous on $[a, b]$ and F is any antiderivative of f (such that $F' = f$), then $\int_a^b f(x) dx = F(b) - F(a)$.

Ex. 7

$$\int_{-1}^1 (1-x)^2 x dx$$

Expand the integrand:

$$\int_{-1}^1 (x^3 - 2x^2 + x) dx$$

Recognize that over a symmetric interval $[-a, a]$, all odd functions (like x^3 and x) evaluate to 0. We only need to integrate the even function component:

$$\int_{-1}^1 -2x^2 dx = 2 \int_0^1 -2x^2 dx = 2 \left[-\frac{2x^3}{3} \right]_0^1 = -\frac{4}{3}$$

Ex. 8

$$\int_0^1 x(\sqrt[3]{x} + \sqrt[4]{x}) dx$$

$$\int_0^1 (x^{4/3} + x^{5/4}) dx = \left[\frac{3}{7}x^{7/3} + \frac{4}{9}x^{9/4} \right]_0^1 = \frac{3}{7} + \frac{4}{9} = \frac{55}{63}$$

Ex. 9

$$\int_1^2 \left(\frac{x}{2} - \frac{2}{x} \right) dx = \left[\frac{x^2}{4} - 2 \ln |x| \right]_1^2 = (1 - 2 \ln 2) - \left(\frac{1}{4} - 0 \right) = \frac{3}{4} - 2 \ln 2$$

Ex. 10

$$\int_0^{\pi/4} \sec x \tan x dx = \left[\sec x \right]_0^{\pi/4} = \sqrt{2} - 1$$

Ex. 11

$$\int_0^{\pi/4} \frac{1 + \cos^3 x}{\cos^2 x} dx = \int_0^{\pi/4} (\sec^2 x + \cos x) dx = \left[\tan x + \sin x \right]_0^{\pi/4} = 1 + \frac{\sqrt{2}}{2}$$

Ex. 12

$$\int_0^{\sqrt{3}/2} \frac{1}{\sqrt{1-x^2}} dx = \left[\arcsin x \right]_0^{\sqrt{3}/2} = \frac{\pi}{3}$$

Ex. 13

$$\int_0^{1/\sqrt{3}} \frac{x^2 - 1}{x^4 - 1} dx$$

Factor the denominator as $(x^2 - 1)(x^2 + 1)$ and cancel the common term:

$$\int_0^{1/\sqrt{3}} \frac{1}{x^2 + 1} dx = \left[\arctan x \right]_0^{1/\sqrt{3}} = \frac{\pi}{6}$$

Integrating Absolute Values

1. **Roots:** Solve $f(x) = 0$ to identify partition points.
2. **Split:** Divide $[a, b]$ into sub-intervals at these roots.
3. **Resolve:** Replace $|f(x)|$ with $f(x)$ or $-f(x)$ based on the sign in each interval.
4. **Integrate:** Evaluate the integrals on each sub-interval and sum the results.

Ex. 14

$$\int_0^2 |2x - 1| dx$$

Find the root ($x = 1/2$) and split the integral bounds to remove the absolute value:

$$\int_0^{1/2} (1 - 2x) dx + \int_{1/2}^2 (2x - 1) dx = \left[x - x^2 \right]_0^{1/2} + \left[x^2 - x \right]_{1/2}^2 = \frac{1}{4} + \left(2 - \left(-\frac{1}{4} \right) \right) = \frac{5}{2}$$

Ex. 15

$$\int_{-1}^1 (|x| - 2x) dx$$

Split at the root ($x = 0$):

$$\begin{aligned} \int_{-1}^0 (-x - 2x) dx + \int_0^1 (x - 2x) dx &= \int_{-1}^0 -3x dx + \int_0^1 -x dx \\ &= \left[-\frac{3}{2}x^2 \right]_{-1}^0 + \left[-\frac{1}{2}x^2 \right]_0^1 = \frac{3}{2} - \frac{1}{2} = 1 \end{aligned}$$

Ex. 16

$$\int_0^{3\pi/2} \sqrt{\sin^2 x} dx$$

A common mistake is evaluating this as simply $\int \sin x dx$. The square root of a square introduces an absolute value:

$$\int_0^{3\pi/2} |\sin x| dx$$

Split the bounds where $\sin x$ becomes negative (at $x = \pi$):

$$\int_0^{\pi} \sin x dx + \int_{\pi}^{3\pi/2} (-\sin x) dx = \left[-\cos x \right]_0^{\pi} + \left[\cos x \right]_{\pi}^{3\pi/2} = (1 - (-1)) + (0 - (-1)) = 2 + 1 = 3$$

Ex. 17

$$\int_0^3 |x^2 - 4| dx$$

Identify the root at $x = 2$ and split the domain:

$$\int_0^2 (4 - x^2) dx + \int_2^3 (x^2 - 4) dx = \left[4x - \frac{x^3}{3}\right]_0^2 + \left[\frac{x^3}{3} - 4x\right]_2^3 = \frac{23}{3}$$

Ex. 18

$$\int_0^4 |\sqrt{x} - 1| dx$$

Identify the root at $x = 1$ and split:

$$\int_0^1 (1 - \sqrt{x}) dx + \int_1^4 (\sqrt{x} - 1) dx = \left[x - \frac{2}{3}x^{3/2}\right]_0^1 + \left[\frac{2}{3}x^{3/2} - x\right]_1^4 = 2$$

Chapter 2

U-Substitution

This section focuses on mastering the speed and mental detection of the core components of u -substitution: spotting the inner function u and its differential du .

The U-Substitution Rule

$$\int f(g(x))g'(x) dx = \int f(u) du = F(u) + C$$

where

$$u = g(x) \quad \text{and} \quad du = g'(x) dx$$

Ex. 1

$$\int x e^{-x^2} dx$$

$$\text{Let } u = -x^2 \implies du = -2x dx \implies x dx = -\frac{1}{2} du.$$

$$\int -\frac{1}{2} e^u du = -\frac{1}{2} e^{-x^2} + C$$

Ex. 2

$$\int \frac{x^3}{x^4 - 5} dx$$

$$\text{Let } u = x^4 - 5 \implies du = 4x^3 dx.$$

$$\frac{1}{4} \int \frac{1}{u} du = \frac{1}{4} \ln |u| + C = \frac{1}{4} \ln |x^4 - 5| + C$$

Ex. 3

$$\int \sqrt{2x - 1} dx$$

$$\text{Let } u = 2x - 1 \implies du = 2 dx.$$

$$\frac{1}{2} \int \sqrt{u} du = \frac{1}{2} \left(\frac{2}{3} u^{3/2} \right) + C = \frac{1}{3} (2x - 1)^{3/2} + C$$

Ex. 4

$$\int (1 - x)^9 dx$$

$$\text{Let } u = 1 - x \implies du = -dx.$$

$$-\int u^9 du = -\frac{u^{10}}{10} + C = -\frac{(1 - x)^{10}}{10} + C$$

Ex. 5

$$\int \frac{1}{5-3x} dx$$

Let $u = 5 - 3x \implies du = -3 dx$.

$$-\frac{1}{3} \int \frac{1}{u} du = -\frac{1}{3} \ln |5 - 3x| + C$$

Ex. 6

$$\int \cos^3 x \sin x dx$$

Let $u = \cos x \implies du = -\sin x dx$.

$$-\int u^3 du = -\frac{u^4}{4} + C = -\frac{\cos^4 x}{4} + C$$

Ex. 7

$$\int \frac{e^x}{1 - e^x} dx$$

Let $u = 1 - e^x \implies du = -e^x dx$.

$$-\int \frac{1}{u} du = -\ln |1 - e^x| + C$$

Ex. 8

$$\int \frac{\sin(\sqrt{x})}{\sqrt{x}} dx$$

Let $u = \sqrt{x} \implies du = \frac{1}{2\sqrt{x}} dx$.

$$2 \int \sin(u) du = -2 \cos(u) + C = -2 \cos(\sqrt{x}) + C$$

Ex. 9

$$\int \frac{\ln^2 x}{x} dx$$

Let $u = \ln x \implies du = \frac{1}{x} dx$.

$$\int u^2 du = \frac{u^3}{3} + C = \frac{\ln^3 x}{3} + C$$

Ex. 10

$$\int \sin(\cos x) \sin x dx$$

Let $u = \cos x \implies du = -\sin x dx$.

$$-\int \sin u du = \cos(u) + C = \cos(\cos x) + C$$

Ex. 11

$$\int x \sqrt{x+2} dx$$

Let $u = x + 2 \implies du = dx$. We still have an x remaining. We use the back-substitution $x = u - 2$.

$$\begin{aligned} \int (u-2)\sqrt{u} du &= \int (u^{3/2} - 2u^{1/2}) du \\ &= \frac{2}{5}u^{5/2} - \frac{4}{3}u^{3/2} + C = \frac{2}{5}(x+2)^{5/2} - \frac{4}{3}(x+2)^{3/2} + C \end{aligned}$$

Ex. 12

$$\int (x^2 + 1)(x^3 + 3x)^4 dx$$

$$\text{Let } u = x^3 + 3x \implies du = (3x^2 + 3) dx = 3(x^2 + 1) dx.$$

$$\frac{1}{3} \int u^4 du = \frac{u^5}{15} + C = \frac{(x^3 + 3x)^5}{15} + C$$

Ex. 13

$$\int \frac{\sec^2 x}{\tan^2 x} dx$$

$$\text{Let } u = \tan x \implies du = \sec^2 x dx.$$

$$\int \frac{1}{u^2} du = -\frac{1}{u} + C = -\cot x + C$$

Ex. 14

$$\int \sin(2x - 1) dx$$

$$\text{Let } u = 2x - 1 \implies du = 2 dx.$$

$$\frac{1}{2} \int \sin(u) du = -\frac{1}{2} \cos(2x - 1) + C$$

Ex. 15

$$\int \frac{\csc^2 x}{\sqrt{\cot x}} dx$$

$$\text{Let } u = \cot x \implies du = -\csc^2 x dx.$$

$$-\int \frac{1}{\sqrt{u}} du = -2\sqrt{u} + C = -2\sqrt{\cot x} + C$$

Ex. 16

$$\int \frac{e^x}{e^x + 1} dx$$

$$\text{Let } u = e^x + 1 \implies du = e^x dx.$$

$$\int \frac{1}{u} du = \ln(e^x + 1) + C$$

Ex. 17

$$\int \frac{\sin(2x)}{1 + \cos^2 x} dx$$

Rather than expanding $\sin(2x) = 2 \sin x \cos x$, check the derivative of the denominator directly: let $u = 1 + \cos^2 x \implies du = -2 \cos x \sin x dx = -\sin(2x) dx$.

$$-\int \frac{1}{u} du = -\ln(1 + \cos^2 x) + C$$

Bonus Fact: The derivative of $\sin^2 x$ is exactly $\sin(2x)$.

Ex. 18

$$\int \cot x dx = \int \frac{\cos x}{\sin x} dx$$

$$\text{Let } u = \sin x \implies du = \cos x dx.$$

$$\int \frac{1}{u} du = \ln |\sin x| + C$$

Ex. 19

$$\int \frac{\cos(\ln x)}{x} dx$$

Let $u = \ln x \implies du = \frac{1}{x} dx$.

$$\int \cos(u) du = \sin(u) + C = \sin(\ln x) + C$$

Ex. 20

$$\int \frac{x}{x^4 + 1} dx$$

We cannot use $u = x^4$ since we only have x on top (we would need x^3). Instead, let $u = x^2 \implies du = 2x dx$. The denominator x^4 becomes u^2 .

$$\frac{1}{2} \int \frac{1}{u^2 + 1} du = \frac{1}{2} \arctan(u) + C = \frac{1}{2} \arctan(x^2) + C$$

Ex. 21

$$\int \frac{1+x}{1+x^2} dx$$

Split the fraction into two components:

$$\int \frac{1}{1+x^2} dx + \int \frac{x}{1+x^2} dx$$

The first is standard $\arctan x$. For the second, let $u = 1 + x^2 \implies du = 2x dx$.

$$\arctan x + \frac{1}{2} \ln(1+x^2) + C$$

Ex. 22

$$\int \frac{x^3}{\sqrt{x^2 + 1}} dx$$

Let $u = x^2 + 1 \implies du = 2x dx$. Split x^3 into $x^2 \cdot x$. Back-substitute $x^2 = u - 1$.

$$\begin{aligned} \frac{1}{2} \int \frac{u-1}{\sqrt{u}} du &= \frac{1}{2} \int (u^{1/2} - u^{-1/2}) du \\ &= \frac{1}{2} \left(\frac{2}{3} u^{3/2} - 2u^{1/2} \right) + C = \frac{1}{3} (x^2 + 1)^{3/2} - \sqrt{x^2 + 1} + C \end{aligned}$$

Ex. 23

$$\int_0^1 \frac{1}{\sqrt[3]{1+7x}} dx$$

Let $u = 1 + 7x \implies du = 7 dx$. Transform the bounds: $x = 0 \implies u = 1$ and $x = 1 \implies u = 8$.

$$\frac{1}{7} \int_1^8 u^{-1/3} du = \frac{1}{7} \left[\frac{3}{2} u^{2/3} \right]_1^8 = \frac{3}{14} (8^{2/3} - 1^{2/3}) = \frac{3}{14} (4 - 1) = \frac{9}{14}$$

Ex. 24

$$\int_1^2 \frac{e^{1/x}}{x^2} dx$$

Let $u = \frac{1}{x} \implies du = -\frac{1}{x^2} dx$. Transform bounds: $x = 1 \implies u = 1$ and $x = 2 \implies u = 1/2$. The bounds are reversed ($1 \rightarrow 1/2$), but the negative sign from the differential corrects this:

$$-\int_1^{1/2} e^u du = \int_{1/2}^1 e^u du = [e^u]_{1/2}^1 = e - \sqrt{e}$$

Ex. 25

$$\int_{-\pi/4}^{\pi/4} (x^3 + x^4 \tan x) dx$$

Observe the symmetry of the bounds. Check the function parity: x^3 is odd. The term $x^4 \tan x$ is (even) $\times$ (odd) = odd. Because the entire function is odd over a symmetric interval $[-a, a]$, the integral equates perfectly to:

$$0$$

Ex. 26

$$\int_1^2 x\sqrt{x-1} dx$$

Let $u = x - 1 \implies du = dx$ and $x = u + 1$. Bounds change to $0 \rightarrow 1$.

$$\int_0^1 (u+1)\sqrt{u} du = \int_0^1 (u^{3/2} + u^{1/2}) du = \left[\frac{2}{5}u^{5/2} + \frac{2}{3}u^{3/2} \right]_0^1 = \frac{2}{5} + \frac{2}{3} = \frac{16}{15}$$

Ex. 27

$$\int_{-\pi}^{\pi} x^4 \sin x dx$$

Since x^4 is even and $\sin x$ is odd, the product is an odd function across symmetric boundaries. The result is immediately:

$$0$$

Ex. 28

$$\int_1^e \frac{\sqrt{\ln x}}{x} dx$$

Let $u = \ln x \implies du = \frac{1}{x} dx$. Bounds change from $0 \rightarrow 1$.

$$\int_0^1 \sqrt{u} du = \left[\frac{2}{3}u^{3/2} \right]_0^1 = \frac{2}{3}$$

Ex. 29

$$\int_0^2 (x-1)e^{(x-1)^2} dx$$

Let $u = x - 1 \implies du = dx$. The bounds shift to $-1 \rightarrow 1$.

$$\int_{-1}^1 ue^{u^2} du$$

Observe that ue^{u^2} is an odd function evaluated over $[-1, 1]$. The result is immediately:

$$0$$

Ex. 30

$$\int_0^1 \frac{e^x}{1+e^x} dx$$

Let $u = e^x + 1 \implies du = e^x dx$. Bounds change to $2 \rightarrow e + 1$.

$$\int_2^{e+1} \frac{1}{u} du = \ln(e+1) - \ln(2)$$

Ex. 31

$$\int_0^1 \sqrt{1-x^2} dx$$

Rather than using trigonometric substitution, recognize this as the equation of a semi-circle $y = \sqrt{R^2 - x^2}$ with radius $R = 1$. The bounds 0 to 1 indicate exactly one-quarter of the circle's total area.

$$A = \frac{\pi R^2}{4} = \frac{\pi(1)^2}{4} = \frac{\pi}{4}$$

Ex. 32

$$\int_0^{11} \sqrt{121-x^2} dx$$

Recognize $121 = 11^2$. This is a quarter circle of radius $R = 11$.

$$\frac{\pi(11)^2}{4} = \frac{121\pi}{4}$$

Ex. 33

$$\int_{-2}^2 \sqrt{12-3x^2} dx$$

Factor out the 3 from the internal polynomial:

$$\int_{-2}^2 \sqrt{3(4-x^2)} dx = \sqrt{3} \int_{-2}^2 \sqrt{4-x^2} dx$$

This reveals a semi-circle of radius $R = 2$ evaluated fully from -2 to 2 , forming a half-circle.

$$\sqrt{3} \left(\frac{\pi(2)^2}{2} \right) = 2\pi\sqrt{3}$$

Ex. 34

$$\int_{-2}^2 (x+1)\sqrt{4-x^2} dx$$

Distribute the radical:

$$\int_{-2}^2 x\sqrt{4-x^2} dx + \int_{-2}^2 \sqrt{4-x^2} dx$$

The first integral is an odd function evaluated symmetrically, yielding 0. The second integral represents a perfect half-circle with radius $R = 2$.

$$0 + \frac{\pi(2)^2}{2} = 2\pi$$

Ex. 35

$$\int_0^1 x\sqrt{1-x^4} dx$$

Let $u = x^2 \implies du = 2x dx$. The bounds $0 \rightarrow 1$ remain the same. The term x^4 becomes u^2 .

$$\frac{1}{2} \int_0^1 \sqrt{1-u^2} du$$

The remaining integral is exactly a quarter-circle with radius $R = 1$.

$$\frac{1}{2} \left(\frac{\pi(1)^2}{4} \right) = \frac{\pi}{8}$$

Ex. 36

$$\int_{-\pi/4}^{\pi/4} \frac{x^4 \tan x}{2 + \cos x} dx$$

Check for parity. x^4 is even. $\tan x$ is odd. The denominator $2 + \cos x$ is even. The complete product is an odd function over symmetric bounds. The answer is directly:

$$0$$

Ex. 37

$$\int (1 + x^{1/4})^2 \frac{1}{\sqrt{x}} dx$$

Expand the perfect square: $(1 + 2x^{1/4} + x^{1/2})$. Multiply by $x^{-1/2}$:

$$\int (x^{-1/2} + 2x^{-1/4} + 1) dx = 2x^{1/2} + \frac{8}{3}x^{3/4} + x + C$$

Ex. 38

$$\int_0^\infty \frac{e^x}{1 + e^{2x}} dx$$

Let $u = e^x \implies du = e^x dx$. The bounds change from $0 \rightarrow \infty$ to $1 \rightarrow \infty$.

$$\int_1^\infty \frac{1}{1 + u^2} du = \left[\arctan(u) \right]_1^\infty = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$$

Ex. 39

$$\int \frac{x + 2}{\sqrt{x^2 + 4x}} dx$$

Let $u = x^2 + 4x \implies du = (2x + 4) dx = 2(x + 2) dx$.

$$\frac{1}{2} \int \frac{1}{\sqrt{u}} du = \sqrt{u} + C = \sqrt{x^2 + 4x} + C$$

Ex. 40

$$\int \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$$

Let $u = \sqrt{x} \implies du = \frac{1}{2\sqrt{x}} dx$.

$$2 \int e^u du = 2e^u + C = 2e^{\sqrt{x}} + C$$

Ex. 41

$$\int \frac{\sec x \tan x}{\sec x + 1} dx$$

Let $u = \sec x + 1 \implies du = \sec x \tan x dx$.

$$\int \frac{1}{u} du = \ln |\sec x + 1| + C$$

Chapter 3

Integration by Parts

This section focuses on mastering the foundational rules of integration by parts, including the DI (Tabular) method, reverse product rule, recursive integrals, and forced U-substitution. Mastery of these fundamentals is necessary before advancing to competitive speed integration techniques.

3.1 The DI (Tabular) Method

Example 1

$$\int x e^{2x} dx$$

Apply the DI method by differentiating x and integrating e^{2x} :

$$\int x e^{2x} dx = \frac{1}{2} x e^{2x} - \frac{1}{4} e^{2x} + C$$

Example 2

$$\int \sqrt{x} \ln(x) dx$$

Apply the DI method by differentiating $\ln(x)$ and integrating $\sqrt{x}$:

$$\int \sqrt{x} \ln(x) dx = \frac{2}{3} x^{3/2} \ln(x) - \frac{4}{9} x^{3/2} + C$$

Example 3

$$\int x^2 \cos(2x) dx$$

Apply the DI method by differentiating x^2 and integrating $\cos(2x)$:

$$\int x^2 \cos(2x) dx = \frac{1}{2} x^2 \sin(2x) + \frac{1}{2} x \cos(2x) - \frac{1}{4} \sin(2x) + C$$

Example 4

$$\int (x^2 + 2x) e^x dx$$

Apply the DI method, observing that the trailing terms perfectly cancel out:

$$\int (x^2 + 2x) e^x dx = x^2 e^x + C$$

3.2 Reverse Product Rule

Example 5

$$\int (5x^4 e^x + x^5 e^x) dx$$

Recognize the integrand as the exact derivative of a product:

$$\int (5x^4 e^x + x^5 e^x) dx = x^5 e^x + C$$

Example 6

$$\int (e^x \sin x + e^x \cos x) dx$$

Recognize the integrand as the exact derivative of $e^x \sin x$:

$$\int (e^x \sin x + e^x \cos x) dx = e^x \sin x + C$$

Example 7

$$\int (2x \sin x + x^2 \cos x) dx$$

Recognize the integrand as the exact derivative of $x^2 \sin x$:

$$\int (2x \sin x + x^2 \cos x) dx = x^2 \sin x + C$$

3.3 Definite Integrals and Algebraic Manipulations

Example 8

$$\int_0^1 \frac{x^2 + 1}{e^x} dx$$

Rewrite with a negative exponent and apply the DI method:

$$\int_0^1 (x^2 + 1)e^{-x} dx = 3 - \frac{6}{e}$$

3.4 Inverse Functions and Logarithms

Example 9

$$\int \arccos(x) dx$$

Apply integration by parts by differentiating $\arccos(x)$ and integrating 1:

$$\int \arccos(x) dx = x \arccos(x) - \sqrt{1 - x^2} + C$$

Example 10

$$\int \ln(\sqrt{x}) dx$$

Simplify using logarithm properties before integrating:

$$\int \ln(\sqrt{x}) dx = \frac{1}{2}x \ln(x) - \frac{1}{2}x + C$$

Example 11

$$\int x^4 \ln(x) dx$$

Apply the DI method by differentiating $\ln(x)$ and integrating x^4 :

$$\int x^4 \ln(x) dx = \frac{x^5}{5} \ln(x) - \frac{x^5}{25} + C$$

Example 12

$$\int \arctan(x) dx$$

Apply integration by parts by differentiating $\arctan(x)$ and integrating 1:

$$\int \arctan(x) dx = x \arctan(x) - \frac{1}{2} \ln(1 + x^2) + C$$

Example 13

$$\int x \csc^2(x) dx$$

Apply the DI method by differentiating x and integrating $\csc^2(x)$:

$$\int x \csc^2(x) dx = -x \cot(x) + \ln |\sin(x)| + C$$

3.5 Forced U-Substitution**Example 14**

$$\int (\ln x)^4 dx$$

Force the substitution $u = \ln x$, yielding $dx = e^u du$:

$$\int (\ln x)^4 dx = x ((\ln x)^4 - 4(\ln x)^3 + 12(\ln x)^2 - 24 \ln x + 24) + C$$

Example 15

$$\int_0^1 (\arcsin x)^2 dx$$

Force the substitution $u = \arcsin x$ to change the bounds and integrand:

$$\int_0^1 (\arcsin x)^2 dx = \frac{\pi^2}{4} - 2$$

Example 16

$$\int e^{\sqrt{x}} dx$$

Force the substitution $u = \sqrt{x}$, yielding $dx = 2u du$:

$$\int e^{\sqrt{x}} dx = 2\sqrt{x}e^{\sqrt{x}} - 2e^{\sqrt{x}} + C$$

Example 17

$$\int \cos(\ln x) dx$$

Force the substitution $u = \ln x$, yielding $dx = e^u du$:

$$\int \cos(\ln x) dx = \frac{x}{2} (\sin(\ln x) + \cos(\ln x)) + C$$

Example 18

$$\int \sin(\sqrt{x}) dx$$

Force the substitution $u = \sqrt{x}$, yielding $dx = 2u du$:

$$\int \sin(\sqrt{x}) dx = -2\sqrt{x} \cos(\sqrt{x}) + 2 \sin(\sqrt{x}) + C$$

3.6 Recursive Integration and Advanced Techniques**Example 19**

$$\int e^{2x} \sin(3x) dx$$

Apply the DI method twice, set equal to I , and solve algebraically:

$$\int e^{2x} \sin(3x) dx = \frac{e^{2x}}{13} (2 \sin(3x) - 3 \cos(3x)) + C$$

Example 20

$$\int x \sin x \cos x dx$$

Convert to $\frac{1}{2} \sin(2x)$ before applying the DI method:

$$\int x \sin x \cos x dx = \frac{1}{8} \sin(2x) - \frac{x}{4} \cos(2x) + C$$

Example 21

$$\int_1^{\sqrt{3}} \arctan\left(\frac{1}{x}\right) dx$$

Apply the DI method by differentiating $\arctan(1/x)$ and integrating 1:

$$\int_1^{\sqrt{3}} \arctan\left(\frac{1}{x}\right) dx = \frac{\pi\sqrt{3}}{6} + \frac{1}{2} \ln(2) - \frac{\pi}{4}$$

Example 22

$$\int_0^{\pi/3} \sin x \ln(\cos x) dx$$

Apply the DI method by differentiating $\ln(\cos x)$ and integrating $\sin x$:

$$\int_0^{\pi/3} \sin x \ln(\cos x) dx = \frac{1}{2} \ln(2) - \frac{1}{2}$$

Example 23: Zero Substitution

$$\int x \ln(x+1) dx$$

Use the zero substitution trick by adding and subtracting 1 and x to divide the polynomial smoothly:

$$\int x \ln(x+1) dx = \frac{x^2}{2} \ln(x+1) - \frac{1}{2} \left(\frac{x^2}{2} - x + \ln(x+1) \right) + C$$

Chapter 4

Trigonometric Integrals

This section focuses on manipulating trigonometric integrals using identities, U-substitutions, Wallis's product, product-to-sum formulas, and recognizing symmetry. Mastery of these techniques is essential for bypassing lengthy algebraic methods during speed integration.

4.1 Sines, Cosines, and U-Substitutions

Example 1

$$\int \sin^2(x) \cos^3(x) dx$$

Factor out one $\cos(x)$ and convert the remaining $\cos^2(x)$ to $1 - \sin^2(x)$, then use the substitution $u = \sin(x)$:

$$\int \sin^2(x)(1 - \sin^2(x)) \cos(x) dx = \int (u^2 - u^4) du = \frac{\sin^3(x)}{3} - \frac{\sin^5(x)}{5} + C$$

Example 2: Wallis's Product (Odd Power)

$$\int_0^{\pi/2} \sin^5(x) dx$$

Using Wallis's trick for odd powers, continuously multiply the sequence of fractions starting from the bottom up to the power n :

$$\int_0^{\pi/2} \sin^5(x) dx = \frac{2 \cdot 4}{3 \cdot 5} = \frac{8}{15}$$

Example 3

$$\int x \cos^5(x^2) dx$$

Let $u = x^2$, making $du = 2x dx$. The integral becomes $\frac{1}{2} \int \cos^5(u) du$. Factor out $\cos(u)$, convert $\cos^4(u)$ to $(1 - \sin^2(u))^2$, and substitute $v = \sin(u)$:

$$\frac{1}{2} \int (1 - v^2)^2 dv = \frac{1}{2} \left(v - \frac{2v^3}{3} + \frac{v^5}{5} \right) = \frac{\sin(x^2)}{2} - \frac{\sin^3(x^2)}{3} + \frac{\sin^5(x^2)}{10} + C$$

Example 4

$$\int \sin^2(x) dx$$

Apply the half-angle identity $\sin^2(x) = \frac{1 - \cos(2x)}{2}$:

$$\int \frac{1 - \cos(2x)}{2} dx = \frac{x}{2} - \frac{\sin(2x)}{4} + C$$

Example 5

$$\int \cos^4(x) dx$$

Apply the half-angle identity twice. First, convert $\cos^2(x)$ to $\frac{1 + \cos(2x)}{2}$, and square it. Apply the identity again to the resulting $\cos^2(2x)$:

$$\int \left(\frac{1}{4} + \frac{\cos(2x)}{2} + \frac{1 + \cos(4x)}{8} \right) dx = \frac{3x}{8} + \frac{\sin(2x)}{4} + \frac{\sin(4x)}{32} + C$$

Example 6

$$\int \sin^2(x) \cos^2(x) dx$$

Recognize $(\sin(x) \cos(x))^2 = \frac{\sin^2(2x)}{4}$. Apply the half-angle identity to $\sin^2(2x)$:

$$\int \frac{1 - \cos(4x)}{8} dx = \frac{x}{8} - \frac{\sin(4x)}{32} + C$$

Example 7

$$\int \sin^3(x) \cos^5(x) dx$$

Factor out $\cos(x)$, substitute $u = \sin(x)$, and convert the remaining $\cos^4(x)$ to $(1 - u^2)^2$:

$$\int u^3(1 - u^2)^2 du = \int (u^3 - 2u^5 + u^7) du = \frac{\sin^4(x)}{4} - \frac{\sin^6(x)}{3} + \frac{\sin^8(x)}{8} + C$$

Example 8

$$\int_0^{\pi/2} (2 - \sin x)^2 dx$$

Expand to $4 - 4\sin(x) + \sin^2(x)$. Integrate each part, using Wallis's trick for the $\sin^2(x)$ term:

$$[4x + 4\cos(x)]_0^{\pi/2} + \frac{\pi}{4} = (2\pi - 0) + (0 - 4) + \frac{\pi}{4} = \frac{9\pi}{4} - 4$$

4.2 Secants and Tangents**Example 9**

$$\int \sec^3(x) \tan(x) dx$$

Let $u = \sec(x)$, so $du = \sec(x) \tan(x) dx$. The integral reduces smoothly:

$$\int u^2 du = \frac{\sec^3(x)}{3} + C$$

Example 10

$$\int \sec^6(x) \tan^4(x) dx$$

Factor out $\sec^2(x) dx$, convert the remaining $\sec^4(x)$ to $(\tan^2(x)+1)^2$, and substitute $u = \tan(x)$:

$$\int u^4(u^2 + 1)^2 du = \int (u^8 + 2u^6 + u^4) du = \frac{\tan^9(x)}{9} + \frac{2 \tan^7(x)}{7} + \frac{\tan^5(x)}{5} + C$$

Example 11

$$\int \tan^3(x) \sec(x) dx$$

Factor out $\sec(x) \tan(x) dx$, convert the remaining $\tan^2(x)$ to $\sec^2(x) - 1$, and let $u = \sec(x)$:

$$\int (u^2 - 1) du = \frac{\sec^3(x)}{3} - \sec(x) + C$$

Example 12

$$\int \tan^2(x) \cos^3(x) dx$$

Convert everything to sines and cosines to find $\frac{\sin^2(x)}{\cos^2(x)} \cos^3(x) = \sin^2(x) \cos(x)$. Let $u = \sin(x)$:

$$\int u^2 du = \frac{\sin^3(x)}{3} + C$$

Example 13

$$\int (\tan^2(x) + \tan^4(x)) dx$$

Factor out $\tan^2(x)$ to get $\tan^2(x)(1 + \tan^2(x)) = \tan^2(x) \sec^2(x)$. Let $u = \tan(x)$:

$$\int u^2 du = \frac{\tan^3(x)}{3} + C$$

Example 14

$$\int \tan^4(x) dx$$

Convert a $\tan^2(x)$ to $\sec^2(x) - 1$. Expand and apply the identity again on the remaining squared tangent:

$$\int (\tan^2(x) \sec^2(x) - \sec^2(x) + 1) dx = \frac{\tan^3(x)}{3} - \tan(x) + x + C$$

Example 15

$$\int \tan^5(x) dx$$

Peel off a factor of $\tan(x) \sec^2(x)$ and repeatedly apply $\tan^2(x) = \sec^2(x) - 1$:

$$\int (\tan^3(x) \sec^2(x) - \tan^3(x)) dx = \frac{\tan^4(x)}{4} - \frac{\tan^2(x)}{2} - \ln |\cos(x)| + C$$

4.3 Product-to-Sum and Advanced Manipulations

Example 16

$$\int \sin(4x) \cos(5x) dx$$

Apply the product-to-sum identity $\sin(A) \cos(B) = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$:

$$\frac{1}{2} \int (\sin(9x) - \sin(x)) dx = -\frac{\cos(9x)}{18} + \frac{\cos(x)}{2} + C$$

Example 17

$$\int \cos(2x) \cos(3x) dx$$

Apply the product-to-sum identity $\cos(A) \cos(B) = \frac{1}{2}[\cos(A+B) + \cos(A-B)]$:

$$\frac{1}{2} \int (\cos(5x) + \cos(x)) dx = \frac{\sin(5x)}{10} + \frac{\sin(x)}{2} + C$$

Example 18

$$\int \sin(6x) \sin(8x) dx$$

Apply the product-to-sum identity $\sin(A) \sin(B) = \frac{1}{2}[\cos(A-B) - \cos(A+B)]$:

$$\frac{1}{2} \int (\cos(2x) - \cos(14x)) dx = \frac{\sin(2x)}{4} - \frac{\sin(14x)}{28} + C$$

Example 19

$$\int_0^{\pi/6} \sqrt{1 + \cos(2x)} dx$$

Use the identity $1 + \cos(2x) = 2 \cos^2(x)$. Because of the square root, use absolute value $|\cos(x)|$:

$$\int_0^{\pi/6} \sqrt{2} |\cos(x)| dx = \sqrt{2} [\sin(x)]_0^{\pi/6} = \frac{\sqrt{2}}{2}$$

Example 20

$$\int \frac{1 - \tan^2(x)}{1 + \tan^2(x)} dx$$

Recognize the denominator is $\sec^2(x)$. Convert completely to sines and cosines: $\cos^2(x) - \sin^2(x) = \cos(2x)$:

$$\int \cos(2x) dx = \frac{\sin(2x)}{2} + C$$

Example 21: Trig Conjugation

$$\int \frac{1}{1 - \cos(x)} dx$$

Multiply the numerator and denominator by the conjugate $1 + \cos(x)$. The denominator becomes $1 - \cos^2(x) = \sin^2(x)$:

$$\int \frac{1 + \cos(x)}{\sin^2(x)} dx = \int (\csc^2(x) + \cot(x) \csc(x)) dx = -\cot(x) - \csc(x) + C$$

4.4 Symmetry Traps and Memorization

Example 22

$$\int_{-\pi/2}^{\pi/2} \sin(2x) \cos(x) dx$$

Before attempting any product-to-sum identities, check the bounds. The integrand is an odd function over symmetric bounds, so the integral evaluates to zero:

$$\int_{-\pi/2}^{\pi/2} \sin(2x) \cos(x) dx = 0$$

Example 23

$$\int \sec^3(x) dx$$

This standard recursive integration by parts problem should be solved purely by memory in a speed competition:

$$\int \sec^3(x) dx = \frac{1}{2}(\sec(x) \tan(x) + \ln |\sec(x) + \tan(x)|) + C$$

Example 24: Symmetry Trap (Odd Power)

$$\int_{-\pi}^{\pi} \sin^3(x) dx$$

Do not default to expanding the integral. Always notice symmetric bounds combined with an odd function:

$$\int_{-\pi}^{\pi} \sin^3(x) dx = 0$$

Chapter 5

Trig-Substitution

This summary covers the examples from the Trigonometric Substitution chapter, demonstrating methods for simplifying radical integrands using standard identities or alternative algebraic substitutions.

5.1 Examples and Solutions

Example 1: Basic Sine Substitution

Evaluate the indefinite integral:

$$\int \frac{x^2}{\sqrt{1-x^2}} dx$$

Let $x = \sin \theta$, which implies $dx = \cos \theta d\theta$ and $\theta = \sin^{-1}(x)$.

$$\begin{aligned} \int \frac{x^2}{\sqrt{1-x^2}} dx &= \int \frac{\sin^2 \theta}{\cos \theta} \cdot \cos \theta d\theta \\ &= \int \sin^2 \theta d\theta \\ &= \int \left(\frac{1}{2} - \frac{1}{2} \cos(2\theta) \right) d\theta \\ &= \frac{\theta}{2} - \frac{1}{4} \sin(2\theta) + C \\ &= \frac{1}{2} \theta - \frac{1}{2} \sin \theta \cos \theta + C \end{aligned}$$

Using the reference triangle where $\sin \theta = x$ and $\cos \theta = \sqrt{1-x^2}$, we get:

$$\frac{1}{2} \sin^{-1}(x) - \frac{1}{2} x \sqrt{1-x^2} + C$$

Example 2: Secant Substitution with Powers

Evaluate the indefinite integral:

$$\int \frac{\sqrt{x^2-1}}{x^4} dx$$

Let $x = \sec \theta$, which implies $dx = \sec \theta \tan \theta d\theta$.

$$\begin{aligned} \int \frac{\tan \theta}{\sec^4 \theta} \cdot \sec \theta \tan \theta d\theta &= \int \frac{\tan^2 \theta}{\sec^3 \theta} d\theta \\ &= \int \sin^2 \theta \cos \theta d\theta \end{aligned}$$

Let $u = \sin \theta$, so $du = \cos \theta d\theta$:

$$\int u^2 du = \frac{1}{3}u^3 + C = \frac{1}{3}\sin^3 \theta + C$$

From the reference triangle where $\sec \theta = x$, we have $\sin \theta = \frac{\sqrt{x^2-1}}{x}$. Substituting back gives:

$$\frac{(x^2-1)^{3/2}}{3x^3} + C$$

Example 3: Direct Algebraic u -Substitution Alternative

$$\int_0^6 \frac{x}{\sqrt{36-x^2}} dx$$

Instead of trigonometric substitution, an algebraic substitution works more directly. Let $u = 36 - x^2$, so $du = -2x dx \implies x dx = -\frac{1}{2}du$. Changing the limits: when $x = 0$, $u = 36$; when $x = 6$, $u = 0$.

$$\int_{36}^0 -\frac{1}{2} \frac{du}{\sqrt{u}} = \int_0^{36} \frac{du}{2\sqrt{u}}$$

Evaluating the foundational radical integral:

$$\left[\sqrt{u} \right]_0^{36} = \sqrt{36} - 0 = 6$$

Example 4: Secant Substitution with Simplification

Evaluate the indefinite integral:

$$\int \frac{dx}{x^2 \sqrt{x^2-1}}$$

Let $x = \sec \theta$, so $dx = \sec \theta \tan \theta d\theta$.

$$\int \frac{\sec \theta \tan \theta d\theta}{\sec^2 \theta \tan \theta} = \int \frac{1}{\sec \theta} d\theta = \int \cos \theta d\theta = \sin \theta + C$$

Using the reference triangle where $\sec \theta = x$, we have $\sin \theta = \frac{\sqrt{x^2-1}}{x}$:

$$\frac{\sqrt{x^2-1}}{x} + C$$

Example 5: Higher-Power Radical Identity

Evaluate the indefinite integral:

$$\int \frac{dx}{(x^2-1)^{3/2}}$$

Let $x = \sec \theta$, so $dx = \sec \theta \tan \theta d\theta$ and $\sqrt{x^2-1} = \tan \theta$.

$$\int \frac{\sec \theta \tan \theta d\theta}{\tan^3 \theta} = \int \frac{\sec \theta}{\tan^2 \theta} d\theta = \int \frac{\cos \theta}{\sin^2 \theta} d\theta$$

Let $u = \sin \theta$, so $du = \cos \theta d\theta$:

$$\int u^{-2} du = -\frac{1}{u} + C = -\csc \theta + C$$

Using the reference triangle where $\csc \theta = \frac{x}{\sqrt{x^2-1}}$:

$$-\frac{x}{\sqrt{x^2-1}} + C$$

Example 6: Scaling and Geometric Interpretation

$$\int_0^{2/3} \sqrt{4 - 9x^2} dx$$

Factor out the constant from the radical to match a standard circular curve form:

$$\int_0^{2/3} \sqrt{9 \left(\frac{4}{9} - x^2 \right)} dx = 3 \int_0^{2/3} \sqrt{\frac{4}{9} - x^2} dx$$

The integral $\int_0^{2/3} \sqrt{\frac{4}{9} - x^2} dx$ represents the area under a quarter circle centered at the origin with radius $r = \frac{2}{3}$ from $x = 0$ to $x = \frac{2}{3}$.

$$3 \cdot \left(\frac{1}{4} \pi r^2 \right) = 3 \cdot \left(\frac{\pi \left(\frac{4}{9} \right)}{4} \right) = \frac{\pi}{3}$$

Example 7: Basic Tangent Substitution

Evaluate the indefinite integral:

$$\int \frac{dx}{\sqrt{4 + x^2}}$$

Let $x = 2 \tan \theta$, which means $\tan \theta = \frac{x}{2}$ and $dx = 2 \sec^2 \theta d\theta$. From the reference triangle, $\sec \theta = \frac{\sqrt{4+x^2}}{2}$.

$$\begin{aligned} \int \frac{2 \sec^2 \theta d\theta}{2 \sec \theta} &= \int \sec \theta d\theta \\ &= \ln |\sec \theta + \tan \theta| + C \\ &= \ln \left| \frac{\sqrt{4+x^2}}{2} + \frac{x}{2} \right| + C \\ &= \ln \left| \sqrt{4+x^2} + x \right| + C_1 \end{aligned}$$

Example 8: Definite Secant Substitution with Powers

$$\int_1^{\sqrt{2}} \frac{\sqrt{x^2 - 1}}{x^3} dx$$

Let $x = \sec \theta$, so $dx = \sec \theta \tan \theta d\theta$. Change the limits: when $x = 1$, $\theta = 0$; when $x = \sqrt{2}$, $\theta = \frac{\pi}{4}$.

$$\begin{aligned} \int_0^{\pi/4} \frac{\tan \theta \cdot \sec \theta \tan \theta d\theta}{\sec^3 \theta} &= \int_0^{\pi/4} \frac{\tan^2 \theta}{\sec^2 \theta} d\theta \\ &= \int_0^{\pi/4} \sin^2 \theta d\theta \\ &= \int_0^{\pi/4} \left(\frac{1}{2} - \frac{1}{2} \cos(2\theta) \right) d\theta \\ &= \left[\frac{\theta}{2} - \frac{1}{4} \sin(2\theta) \right]_0^{\pi/4} \\ &= \frac{\pi}{8} - \frac{1}{4} \end{aligned}$$

Example 9: Definite Integration with Power of Tangent Form

$$\int_0^1 \frac{dx}{(1+x^2)^2}$$

Let $x = \tan \theta$, so $dx = \sec^2 \theta d\theta$. Change the limits: when $x = 0$, $\theta = 0$; when $x = 1$, $\theta = \frac{\pi}{4}$.

$$\begin{aligned} \int_0^{\pi/4} \frac{\sec^2 \theta d\theta}{\sec^4 \theta} &= \int_0^{\pi/4} \cos^2 \theta d\theta \\ &= \int_0^{\pi/4} \left(\frac{1}{2} + \frac{1}{2} \cos(2\theta) \right) d\theta \\ &= \left[\frac{\theta}{2} + \frac{1}{4} \sin(2\theta) \right]_0^{\pi/4} \\ &= \frac{\pi}{8} + \frac{1}{4} \end{aligned}$$

Example 10: Rational Tangent Substitution to u -Substitution

Evaluate the indefinite integral:

$$\int \frac{\sqrt{x^2+1}}{x} dx$$

Let $x = \tan \theta$, so $dx = \sec^2 \theta d\theta$.

$$\int \frac{\sec^3 \theta d\theta}{\tan \theta} = \int \frac{\sec^2 \theta}{\tan \theta} \sec \theta d\theta = \int \frac{\sec \theta}{\sin \theta \cos \theta} \cos \theta d\theta = \int \frac{\sec^2 \theta}{\tan \theta} d\theta$$

Alternatively, converting to sines and cosines:

$$\int \frac{1}{\cos^3 \theta} \cdot \frac{\cos \theta}{\sin \theta} d\theta = \int \frac{d\theta}{\sin \theta \cos^2 \theta} = \int \frac{\sin \theta}{\sin^2 \theta \cos^2 \theta} d\theta$$

Let $u = \cos \theta$, so $du = -\sin \theta d\theta$:

$$\int \frac{-du}{(1-u^2)u^2} = \int \left(-\frac{1}{u^2} - \frac{1}{1-u^2} \right) du = \frac{1}{u} - \frac{1}{2} \ln \left| \frac{1+u}{1-u} \right| + C$$

Substituting back $u = \cos \theta = \frac{1}{\sqrt{x^2+1}}$:

$$\sqrt{x^2+1} - \ln \left| \frac{\sqrt{x^2+1}+1}{x} \right| + C$$

Example 11: Completing the Square followed by Tangent Substitution

Evaluate the indefinite integral:

$$\int \frac{dx}{\sqrt{x^2+2x+5}}$$

First, complete the square for the quadratic expression under the radical:

$$x^2 + 2x + 5 = (x+1)^2 + 4$$

Let $x + 1 = 2 \tan \theta$, so $dx = 2 \sec^2 \theta d\theta$, and $\sec \theta = \frac{\sqrt{x^2+2x+5}}{2}$.

$$\begin{aligned} \int \frac{2 \sec^2 \theta d\theta}{2 \sec \theta} &= \int \sec \theta d\theta \\ &= \ln |\sec \theta + \tan \theta| + C \\ &= \ln \left| \frac{\sqrt{x^2+2x+5}}{2} + \frac{x+1}{2} \right| + C \\ &= \ln \left| \sqrt{x^2+2x+5} + x+1 \right| + C_1 \end{aligned}$$

Example 12: Definite Integration by Completing the Square

$$\int_0^1 \sqrt{x-x^2} dx$$

First, complete the square for the quadratic expression under the radical:

$$x-x^2 = -(x^2-x) = -\left(\left(x-\frac{1}{2}\right)^2 - \frac{1}{4}\right) = \frac{1}{4} - \left(x-\frac{1}{2}\right)^2$$

Rewrite the integral:

$$\int_0^1 \sqrt{\frac{1}{4} - \left(x-\frac{1}{2}\right)^2} dx$$

Let $x - \frac{1}{2} = \frac{1}{2} \sin \theta$, which gives $dx = \frac{1}{2} \cos \theta d\theta$.

Determine the new integration limits:

- When $x = 0$, $\frac{1}{2} \sin \theta = -\frac{1}{2} \implies \theta = -\frac{\pi}{2}$
- When $x = 1$, $\frac{1}{2} \sin \theta = \frac{1}{2} \implies \theta = \frac{\pi}{2}$

Substitute these into the integral:

$$\int_{-\pi/2}^{\pi/2} \sqrt{\frac{1}{4} - \frac{1}{4} \sin^2 \theta} \cdot \left(\frac{1}{2} \cos \theta\right) d\theta = \frac{1}{4} \int_{-\pi/2}^{\pi/2} \cos^2 \theta d\theta$$

By symmetry of the integrand:

$$\frac{2}{4} \int_0^{\pi/2} \cos^2 \theta d\theta = \frac{1}{2} \int_0^{\pi/2} \cos^2 \theta d\theta$$

Using the reduction formula $\int_0^{\pi/2} \cos^2 \theta d\theta = \frac{\pi}{4}$:

$$\frac{1}{2} \cdot \frac{\pi}{4} = \frac{\pi}{8}$$

Example 13: Completing the Square to Standard Inverse Sine Form

Evaluate the indefinite integral:

$$\int \frac{dx}{\sqrt{2x-x^2}}$$

Complete the square inside the radical:

$$2x-x^2 = 1-(x-1)^2$$

Substitute this back into the integral:

$$\int \frac{dx}{\sqrt{1 - (x - 1)^2}}$$

Let $u = x - 1$, so $du = dx$:

$$\int \frac{du}{\sqrt{1 - u^2}} = \sin^{-1}(u) + C = \sin^{-1}(x - 1) + C$$

Example 14: Completing the Square followed by Tangent Substitution

Evaluate the indefinite integral:

$$\int \frac{x^2}{(x^2 - 2x + 2)^2} dx$$

Complete the square for the denominator base: $x^2 - 2x + 2 = (x - 1)^2 + 1$.

$$\int \frac{x^2}{[(x - 1)^2 + 1]^2} dx$$

Let $x - 1 = \tan \theta$, which means $x = \tan \theta + 1$ and $dx = \sec^2 \theta d\theta$.

$$\int \frac{(\tan \theta + 1)^2 \sec^2 \theta}{\sec^4 \theta} d\theta = \int \frac{\tan^2 \theta + 2 \tan \theta + 1}{\sec^2 \theta} d\theta$$

Since $\tan^2 \theta + 1 = \sec^2 \theta$:

$$\int \frac{\sec^2 \theta + 2 \tan \theta}{\sec^2 \theta} d\theta = \int (1 + 2 \sin \theta \cos \theta) d\theta = \int (1 + \sin(2\theta)) d\theta$$

Integrating yields:

$$\theta - \frac{1}{2} \cos(2\theta) + C$$

Substituting back $\theta = \tan^{-1}(x - 1)$:

$$\tan^{-1}(x - 1) - \frac{1}{2} \cos(2 \tan^{-1}(x - 1)) + C$$

Example 15: Algebraic Secant Substitution with High Powers

Evaluate the indefinite integral:

$$\int \frac{dx}{x^4 \sqrt{x^2 - 1}}$$

Let $x = \sec \theta$, so $dx = \sec \theta \tan \theta d\theta$.

$$\int \frac{\sec \theta \tan \theta}{\sec^4 \theta \sqrt{\sec^2 \theta - 1}} d\theta = \int \frac{\sec \theta \tan \theta}{\sec^4 \theta \tan \theta} d\theta = \int \cos^3 \theta d\theta$$

Separate the terms: $\cos^3 \theta = \cos \theta (1 - \sin^2 \theta)$. Let $u = \sin \theta$, $du = \cos \theta d\theta$:

$$\int (1 - u^2) du = u - \frac{1}{3} u^3 + C = \sin \theta - \frac{1}{3} \sin^3 \theta + C$$

Substituting back $\theta = \sec^{-1}(x)$:

$$\sin(\sec^{-1}(x)) - \frac{1}{3} \sin^3(\sec^{-1}(x)) + C$$

Example 16: Definite Radical Integration via Scaling and Secant Substitution

$$\int_{3/2}^{3\sqrt{2}/2} \frac{\sqrt{4x^2 - 9}}{x} dx$$

Factor out the constant from the radical:

$$2 \int_{3/2}^{3\sqrt{2}/2} \frac{\sqrt{x^2 - \frac{9}{4}}}{x} dx$$

Let $x = \frac{3}{2} \sec \theta$, so $dx = \frac{3}{2} \sec \theta \tan \theta d\theta$.

- When $x = \frac{3}{2} \implies \sec \theta = 1 \implies \theta = 0$
- When $x = \frac{3\sqrt{2}}{2} \implies \sec \theta = \sqrt{2} \implies \theta = \frac{\pi}{4}$

Substituting into the expression:

$$2 \int_0^{\pi/4} \frac{\frac{3}{2} \tan \theta}{\frac{3}{2} \sec \theta} \cdot \left(\frac{3}{2} \sec \theta \tan \theta \right) d\theta = 3 \int_0^{\pi/4} \tan^2 \theta d\theta$$

Using $\tan^2 \theta = \sec^2 \theta - 1$:

$$3 \left[\tan \theta - \theta \right]_0^{\pi/4} = 3 \left(1 - \frac{\pi}{4} \right) = 3 - \frac{3\pi}{4}$$

Example 17: Rational Rational Integration via u -Substitution

$$\int_0^{1/\sqrt{2}} \frac{x^3}{(1-x^2)^2} dx$$

Let $u = 1 - x^2$, which implies $x^2 = 1 - u$ and $du = -2x dx \implies x dx = -\frac{1}{2} du$.

- When $x = 0 \implies u = 1$
- When $x = \frac{1}{\sqrt{2}} \implies u = 1 - \frac{1}{2} = \frac{1}{2}$

Rewrite the integral:

$$\int_1^{1/2} \frac{1-u}{u^2} \left(-\frac{1}{2} \right) du = \frac{1}{2} \int_{1/2}^1 \left(\frac{1}{u^2} - \frac{1}{u} \right) du = \frac{1}{2} \left[-\frac{1}{u} - \ln |u| \right]_{1/2}^1$$

Evaluating at the boundaries:

$$\frac{1}{2} \left[(-1 - 0) - \left(-2 - \ln \left(\frac{1}{2} \right) \right) \right] = \frac{1}{2} \left(1 + \ln \left(\frac{1}{2} \right) \right) = \frac{1}{2} + \frac{1}{2} \ln \left(\frac{1}{2} \right)$$

Example 18: Trigonometric Substitution for Composite Radical Fractions

$$\int_0^1 \frac{\sqrt{1-x}}{1-\sqrt{x}} dx$$

Let $x = \sin^2 \theta$, so $dx = 2 \sin \theta \cos \theta d\theta$. The limits transform: $x = 0 \rightarrow \theta = 0$ and $x = 1 \rightarrow \theta = \frac{\pi}{2}$.

$$\int_0^{\pi/2} \frac{\cos \theta}{1 - \sin \theta} \cdot (2 \sin \theta \cos \theta) d\theta = \int_0^{\pi/2} \frac{2 \sin \theta (1 - \sin^2 \theta)}{1 - \sin \theta} d\theta$$

Factoring the difference of squares:

$$\int_0^{\pi/2} 2 \sin \theta (1 + \sin \theta) d\theta = \int_0^{\pi/2} (2 \sin \theta + 2 \sin^2 \theta) d\theta$$

Evaluate each integrated part over the intervals:

$$\left[-2 \cos \theta \right]_0^{\pi/2} + \int_0^{\pi/2} (1 - \cos(2\theta)) d\theta = 2 + \left[\theta - \frac{1}{2} \sin(2\theta) \right]_0^{\pi/2} = 2 + \frac{\pi}{2}$$

Example 19: Inverse Cosine Integration via Double Angle Substitution

$$\int_0^1 \cos^{-1}(2x^2 - 1) dx$$

Let $x = \cos \theta$, so $dx = -\sin \theta d\theta$. Limits transform: $x = 0 \rightarrow \theta = \frac{\pi}{2}$ and $x = 1 \rightarrow \theta = 0$.

$$\int_{\pi/2}^0 \cos^{-1}(2 \cos^2 \theta - 1) \cdot (-\sin \theta) d\theta = \int_0^{\pi/2} \cos^{-1}(\cos(2\theta)) \sin \theta d\theta$$

Since $2\theta \in [0, \pi]$ over this domain, $\cos^{-1}(\cos(2\theta)) = 2\theta$:

$$\int_0^{\pi/2} 2\theta \sin \theta d\theta$$

Integrate by parts with $u = 2\theta$ and $dv = \sin \theta d\theta$:

$$\left[-2\theta \cos \theta \right]_0^{\pi/2} - \int_0^{\pi/2} (-2 \cos \theta) d\theta = \left[-2\theta \cos \theta + 2 \sin \theta \right]_0^{\pi/2} = 2$$

Chapter 6

Rational Polynomials

This section focuses on rapidly integrating rational functions using zero substitution, the Heaviside cover-up method, completing the square to force inverse tangents, and algebraic manipulations to bypass tedious standard partial fractions.

6.1 Zero Substitution and Algebraic Manipulation

Example 1

$$\int \frac{x^3}{x-1} dx$$

Use a zero substitution by borrowing a 1: subtract 1 and add 1. This allows factoring via the difference of cubes:

$$\int \frac{x^3 - 1 + 1}{x-1} dx = \int \left(x^2 + x + 1 + \frac{1}{x-1} \right) dx = \frac{x^3}{3} + \frac{x^2}{2} + x + \ln|x-1| + C$$

Example 2

$$\int \frac{3x-2}{x+1} dx$$

Force the numerator to match the denominator by adding and subtracting 5:

$$\int \frac{3x+3-5}{x+1} dx = \int \left(3 - \frac{5}{x+1} \right) dx = 3x - 5 \ln|x+1| + C$$

Example 3

$$\int \frac{2x-2}{x^3-x} dx$$

Factor the denominator and cancel out the common $(x-1)$ term:

$$\int \frac{2(x-1)}{x(x-1)(x+1)} dx = \int \frac{2}{x(x+1)} dx$$

By observation or basic partial fractions:

$$2 \int \left(\frac{1}{x} - \frac{1}{x+1} \right) dx = 2 \ln|x| - 2 \ln|x+1| + C$$

6.2 The Heaviside Cover-Up Method

Example 4

$$\int \frac{x-4}{(x-2)(x-3)} dx$$

Set up partial fractions $\frac{A}{x-2} + \frac{B}{x-3}$. Cover up the root and plug it into the remaining expression: For A (plug $x = 2$): $\frac{2-4}{2-3} = 2$. For B (plug $x = 3$): $\frac{3-4}{3-2} = -1$.

$$\int \left(\frac{2}{x-2} - \frac{1}{x-3} \right) dx = 2 \ln|x-2| - \ln|x-3| + C$$

Example 5

$$\int \frac{1}{(x+1)(x+2)(x+4)} dx$$

Apply the Heaviside method to each root (-1, -2, -4):

$$\int \left(\frac{1/3}{x+1} - \frac{1/2}{x+2} + \frac{1/6}{x+4} \right) dx = \frac{1}{3} \ln|x+1| - \frac{1}{2} \ln|x+2| + \frac{1}{6} \ln|x+4| + C$$

Example 6

$$\int \frac{1}{(x+1)^2(x+2)} dx$$

Set up $\frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{x+2}$. Use Heaviside for the highest powers: $B = 1$ (plugging $x = -1$), $C = 1$ (plugging $x = -2$). Solve for A by picking a convenient x , like $x = 0$, yielding $A = -1$:

$$\int \left(\frac{-1}{x+1} + \frac{1}{(x+1)^2} + \frac{1}{x+2} \right) dx = -\ln|x+1| - \frac{1}{x+1} + \ln|x+2| + C$$

6.3 Completing the Square and Forcing Derivatives

Example 7

$$\int \frac{1}{x^2 + x + 1} dx$$

Complete the square in the denominator: $x^2 + x + 1 = (x + 1/2)^2 + 3/4$:

$$\int \frac{1}{(x + 1/2)^2 + 3/4} dx = \frac{2}{\sqrt{3}} \arctan \left(\frac{2x + 1}{\sqrt{3}} \right) + C$$

Example 8

$$\int \frac{x+4}{x^2 + 2x + 5} dx$$

The derivative of the denominator is $2x + 2$. Rewrite the numerator to isolate this:

$$\frac{1}{2} \int \frac{2x+2}{x^2 + 2x + 5} dx + \int \frac{3}{(x+1)^2 + 4} dx = \frac{1}{2} \ln(x^2 + 2x + 5) + \frac{3}{2} \arctan \left(\frac{x+1}{2} \right) + C$$

Example 9

$$\int \frac{x}{x^2 + 4x + 5} dx$$

The derivative of the denominator is $2x + 4$. Rewrite the numerator to match:

$$\frac{1}{2} \int \frac{2x + 4 - 4}{x^2 + 4x + 5} dx = \frac{1}{2} \ln(x^2 + 4x + 5) - 2 \arctan(x + 2) + C$$

6.4 Advanced Manipulations and Substitutions**Example 10: “Alpha” Substitution for Even Powers**

$$\int \frac{1}{x^4 - 16} dx$$

Instead of a massive partial fraction expansion, let $\alpha = x^2$ strictly for algebraic factoring: $\frac{1}{\alpha^2 - 16} = \frac{1}{(\alpha - 4)(\alpha + 4)}$. Apply Heaviside to get $\frac{1/8}{\alpha - 4} - \frac{1/8}{\alpha + 4}$:

$$\frac{1}{8} \int \left(\frac{1}{x^2 - 4} - \frac{1}{x^2 + 4} \right) dx = -\frac{1}{16} \operatorname{artanh}\left(\frac{x}{2}\right) - \frac{1}{16} \arctan\left(\frac{x}{2}\right) + C$$

Example 11: “Alpha” Substitution (Sum of Squares)

$$\int \frac{1}{1 - x^4} dx$$

Let $\alpha = x^2$, yielding $\frac{1}{(1 - \alpha)(1 + \alpha)} = \frac{1/2}{1 - \alpha} + \frac{1/2}{1 + \alpha}$. Substitute x^2 back in:

$$\frac{1}{2} \int \left(\frac{1}{1 - x^2} + \frac{1}{1 + x^2} \right) dx = \frac{1}{2} \operatorname{artanh}(x) + \frac{1}{2} \arctan(x) + C$$

Example 12

$$\int \frac{2x + 5}{(x^2 + 5x + 4)(x^2 + 5x + 6)} dx$$

Notice the derivative of $x^2 + 5x$ is $2x + 5$. Let $u = x^2 + 5x + 4$, so $du = (2x + 5)dx$. The integral becomes:

$$\int \frac{1}{u(u + 2)} du = \frac{1}{2} \int \left(\frac{1}{u} - \frac{1}{u + 2} \right) du = \frac{1}{2} \ln|x^2 + 5x + 4| - \frac{1}{2} \ln|x^2 + 5x + 6| + C$$

Example 13

$$\int \frac{e^x}{e^{2x} + 3e^x + 2} dx$$

Substitute $u = e^x$, yielding $\frac{1}{u^2 + 3u + 2} = \frac{1}{(u + 1)(u + 2)}$. Apply Heaviside:

$$\int \left(\frac{1}{u + 1} - \frac{1}{u + 2} \right) du = \ln(e^x + 1) - \ln(e^x + 2) + C$$

Example 14

$$\int \frac{\sin x}{\cos^2 x - 3 \cos x} dx$$

Substitute $u = \cos x$, $du = -\sin x dx$:

$$-\int \frac{1}{u(u - 3)} du = \int \left(\frac{1/3}{u} - \frac{1/3}{u - 3} \right) du = \frac{1}{3} \ln|\cos x| - \frac{1}{3} \ln|\cos x - 3| + C$$

6.5 Creative Zero Substitutions

Example 15

$$\int \frac{1}{x(x-2)^2} dx$$

Set up partial fractions $\frac{A}{x} + \frac{B}{x-2} + \frac{C}{(x-2)^2}$. Cover-up gives $A = 1/4$ (at $x = 0$) and $C = 1/2$ (at $x = 2$); matching the x^2 coefficient gives $B = -1/4$:

$$\int \left(\frac{1/4}{x} - \frac{1/4}{x-2} + \frac{1/2}{(x-2)^2} \right) dx = \frac{1}{4} \ln|x| - \frac{1}{4} \ln|x-2| - \frac{1}{2(x-2)} + C$$

Example 16

$$\int \frac{1}{x(x^2+1)} dx$$

Add and subtract x^2 in the numerator to manually split the fraction:

$$\int \frac{1+x^2-x^2}{x(x^2+1)} dx = \int \left(\frac{1}{x} - \frac{x}{x^2+1} \right) dx = \ln|x| - \frac{1}{2} \ln(x^2+1) + C$$

Chapter 7

Strategic Wildcard

This section focuses on compiling various strategies, throwing a "wildcard" of integrals that test your ability to quickly identify the best method—whether it be zero substitution, integration by parts, symmetry traps, or forced U-substitutions.

7.1 Clever Substitutions and Evaluating Constants

Example 1: Intimidating Constants

$$\int (x+1)^{\sqrt{2}} dx$$

Do not let the radical exponent intimidate you; it is strictly a constant. Let $u = x + 1$:

$$\int u^{\sqrt{2}} du = \frac{(x+1)^{\sqrt{2}+1}}{\sqrt{2}+1} + C$$

Example 2

$$\int e^{e^x+x} dx$$

Rewrite the exponent rules to factor the terms $\int e^{e^x} e^x dx$. Let $u = e^x$, $du = e^x dx$:

$$\int e^u du = e^{e^x} + C$$

Example 3

$$\int e^2 dx$$

Recognize immediately that e^2 contains no variable and is simply a constant:

$$\int e^2 dx = xe^2 + C$$

7.2 Algebraic Manipulations and Zero Substitutions

Example 4

$$\int_0^1 \frac{x}{(2x+1)^2} dx$$

Let $u = 2x + 1$, $dx = \frac{1}{2}du$, and $x = \frac{u-1}{2}$. Adjust the bounds from $[0, 1]$ to $[1, 3]$:

$$\frac{1}{4} \int_1^3 \frac{u-1}{u^2} du = \frac{1}{4} \int_1^3 \left(\frac{1}{u} - \frac{1}{u^2} \right) du = \frac{1}{4} \left[\ln|u| + \frac{1}{u} \right]_1^3 = \frac{1}{4} \left(\ln(3) - \frac{2}{3} \right)$$

Example 5

$$\int \frac{1+12x}{1+3x} dx$$

Use zero substitution to match the numerator with the denominator. Multiply the denominator by 4 to get $12x + 4$, so add and subtract 3:

$$\int \frac{4(1+3x) - 3}{1+3x} dx = \int \left(4 - \frac{3}{1+3x} \right) dx = 4x - \ln|1+3x| + C$$

Example 6

$$\int \frac{1}{1+e^x} dx$$

Add and subtract e^x in the numerator to manually split the fraction:

$$\int \frac{1+e^x - e^x}{1+e^x} dx = \int \left(1 - \frac{e^x}{1+e^x} \right) dx = x - \ln(1+e^x) + C$$

Example 7: Sneaky U-Substitution

$$\int \frac{\sec x \tan x}{\sec^2 x - \sec x} dx$$

Let $u = \sec x$, $du = \sec x \tan x dx$. The integral factors nicely:

$$\int \frac{1}{u^2 - u} du = \int \frac{1}{u(u-1)} du = \int \left(\frac{1}{u-1} - \frac{1}{u} \right) du = \ln|\sec x - 1| - \ln|\sec x| + C$$

7.3 Integration by Parts

Example 8

$$\int \frac{\cos(1/x)}{x^3} dx$$

Split $\frac{1}{x^3}$ into $\frac{1}{x} \cdot \frac{1}{x^2}$. Let $u = \frac{1}{x}$, $du = -\frac{1}{x^2} dx$, then use Integration by Parts:

$$-\int u \cos u du = -(u \sin u + \cos u) + C = -\frac{\sin(1/x)}{x} - \cos(1/x) + C$$

Example 9

$$\int x \sec x \tan x dx$$

Use integration by parts by picking $u = x$ and $dv = \sec x \tan x dx$:

$$x \sec x - \int \sec x dx = x \sec x - \ln|\sec x + \tan x| + C$$

Example 10

$$\int \ln(x^2 + 1) dx$$

Use integration by parts with $u = \ln(x^2 + 1)$ and $dv = dx$:

$$x \ln(x^2 + 1) - \int \frac{2x^2}{x^2 + 1} dx = x \ln(x^2 + 1) - 2 \int \left(1 - \frac{1}{x^2 + 1}\right) dx = x \ln(x^2 + 1) - 2x + 2 \arctan(x) + C$$

Example 11

$$\int \frac{\arctan(x)}{x^2} dx$$

Use integration by parts with $u = \arctan(x)$ and $dv = \frac{1}{x^2} dx$, then use partial fractions (or zero sub):

$$-\frac{\arctan(x)}{x} + \int \frac{1}{x(1+x^2)} dx = -\frac{\arctan(x)}{x} + \int \left(\frac{1}{x} - \frac{x}{1+x^2}\right) dx = -\frac{\arctan(x)}{x} + \ln|x| - \frac{1}{2} \ln(1+x^2) + C$$

7.4 Trigonometric Strategies and Symmetry**Example 12**

$$\int \frac{\sin^3 x}{\cos x} dx$$

Convert $\sin^2 x$ to $1 - \cos^2 x$, and substitute $u = \cos x$:

$$\int \frac{(1-u^2)(-du)}{u} = \int \left(u - \frac{1}{u}\right) du = \frac{\cos^2 x}{2} - \ln|\cos x| + C$$

Example 13

$$\int \frac{1 + \sin x}{1 + \cos x} dx$$

Split the fraction. Multiply the first part by the conjugate $1 - \cos x$, and use u -sub for the second:

$$\int \frac{1 - \cos x}{\sin^2 x} dx + \int \frac{\sin x}{1 + \cos x} dx = \int (\csc^2 x - \cot x \csc x) dx - \ln(1 + \cos x) = -\cot x + \csc x - \ln(1 + \cos x) + C$$

Example 14

$$\int \sqrt{\frac{1+x}{1-x}} dx$$

Multiply the numerator and denominator by the conjugate $\sqrt{1+x}$ to clear the root on top:

$$\int \frac{1+x}{\sqrt{1-x^2}} dx = \int \frac{1}{\sqrt{1-x^2}} dx + \int \frac{x}{\sqrt{1-x^2}} dx = \arcsin(x) - \sqrt{1-x^2} + C$$

Example 15: Symmetry Trap

$$\int_{-\pi/2}^{\pi/2} \frac{x}{1 + \cos x} dx$$

Always check symmetric bounds first! Since x is an odd function and $1 + \cos x$ is even, the overall product is odd:

$$\int_{-\pi/2}^{\pi/2} \frac{x}{1 + \cos x} dx = 0$$

7.5 Advanced and Forced U-Substitutions

Example 16

$$\int \arctan(\sqrt{x}) dx$$

Force substitution $u = \sqrt{x}$, yielding $dx = 2u du$, followed by integration by parts:

$$\int 2u \arctan(u) du = u^2 \arctan(u) - \int \frac{u^2}{1+u^2} du = x \arctan(\sqrt{x}) - \sqrt{x} + \arctan(\sqrt{x}) + C$$

Example 17

$$\int (1 + \sqrt{x})^4 dx$$

Avoid expanding the 4th power mechanically! Force substitution $u = 1 + \sqrt{x}$, meaning $x = (u - 1)^2$ and $dx = 2(u - 1) du$:

$$\int u^4 \cdot 2(u - 1) du = 2 \int (u^5 - u^4) du = \frac{(1 + \sqrt{x})^6}{3} - \frac{2(1 + \sqrt{x})^5}{5} + C$$

Example 18: Hidden Fractional Power

$$\int \frac{\sqrt{x}}{1 + x^3} dx$$

Rewrite x^3 as $(x^{3/2})^2$. Let $u = x^{3/2}$, so $du = \frac{3}{2}\sqrt{x} dx$:

$$\frac{2}{3} \int \frac{1}{1 + u^2} du = \frac{2}{3} \arctan(x^{3/2}) + C$$

Example 19: The "Substitute Everything" Trick

$$\int \sqrt{1 + e^x} dx$$

Let u equal the entire root $u = \sqrt{1 + e^x}$, meaning $u^2 = 1 + e^x$ and $2u du = e^x dx = (u^2 - 1)dx$:

$$\int u \frac{2u}{u^2 - 1} du = 2 \int \left(1 + \frac{1}{u^2 - 1}\right) du = 2\sqrt{1 + e^x} - 2 \operatorname{artanh}(\sqrt{1 + e^x}) + C$$

Example 20

$$\int x^5 e^{-x^3} dx$$

Split x^5 into $x^3 \cdot x^2$. Let $u = x^3$, $du = 3x^2 dx$, then use integration by parts:

$$\frac{1}{3} \int u e^{-u} du = \frac{1}{3} (-u e^{-u} - e^{-u}) + C = -\frac{1}{3} e^{-x^3} (x^3 + 1) + C$$

Chapter 8

More Strategic Wildcards

This section continues the exploration of "wildcard" integrals from Chapter 7, focusing on speed integration strategies such as the caveman trick, advanced forced substitutions, algebraic/trigonometric conjugation, and symmetry traps.

8.1 Product Rule and Algebraic Factoring

Example 1: The Product Rule Trap

$$\int \frac{e^x(x-1)}{x^2} dx$$

Instead of using integration by parts, recognize this as a direct application of the quotient/product rule. Splitting the terms inside the integrand yields:

$$\int e^x \left(\frac{1}{x} - \frac{1}{x^2} \right) dx$$

Since $\frac{d}{dx} \left(\frac{1}{x} \right) = -\frac{1}{x^2}$, this perfectly matches the expansion of $\frac{d}{dx} \left(\frac{e^x}{x} \right)$:

$$\int \frac{e^x(x-1)}{x^2} dx = \frac{e^x}{x} + C$$

Example 2: The Caveman Trick

$$\int \frac{1}{x(x^4+1)} dx$$

Avoid tedious partial fractions. Instead, force a u -substitution by factoring out the highest power (x^4) from the denominator:

$$\int \frac{1}{x \cdot x^4(1+x^{-4})} dx = \int \frac{x^{-5}}{1+x^{-4}} dx$$

Let $u = 1 + x^{-4}$, which means $du = -4x^{-5} dx$:

$$-\frac{1}{4} \int \frac{1}{u} du = -\frac{1}{4} \ln(1+x^{-4}) + C$$

Example 3: Power Expansion and Wallis' Formula

$$\int_0^{\pi/2} (x + \sin x)^2 dx$$

Completely expand the binomial expression before evaluating the individual pieces:

$$\int_0^{\pi/2} (x^2 + 2x \sin x + \sin^2 x) dx$$

Evaluating term-by-term:

1. $\int_0^{\pi/2} x^2 dx = \left[\frac{x^3}{3} \right]_0^{\pi/2} = \frac{\pi^3}{24}$
2. $2 \int_0^{\pi/2} x \sin x dx$ evaluated via integration by parts yields $2 [\sin x - x \cos x]_0^{\pi/2} = 2(1 - 0) = 2$
3. $\int_0^{\pi/2} \sin^2 x dx$ can be solved immediately using Wallis' Product rule: $\frac{1}{2} \cdot \frac{\pi}{2} = \frac{\pi}{4}$

Summing the terms together gives the final value:

$$\frac{\pi^3}{24} + 2 + \frac{\pi}{4}$$

Example 4

$$\int \frac{1}{\sqrt{x} + x\sqrt{x}} dx$$

Factor out $\sqrt{x}$ from the denominator to make the underlying structure clearer:

$$\int \frac{1}{\sqrt{x}(1+x)} dx$$

Let $u = \sqrt{x}$, meaning $du = \frac{1}{2\sqrt{x}} dx \implies 2 du = \frac{1}{\sqrt{x}} dx$:

$$\int \frac{2}{1+u^2} du = 2 \arctan(u) + C = 2 \arctan(\sqrt{x}) + C$$

8.2 Trigonometric Conversions and Forced Substitutions

Example 5

$$\int \frac{1}{1 + \cos^2 x} dx$$

There is no direct standard identity for $1 + \cos^2 x$. Multiply the numerator and denominator by $\sec^2 x$ to switch into a coordinate system driven by tangents and secants:

$$\int \frac{\sec^2 x}{\sec^2 x + 1} dx = \int \frac{\sec^2 x}{(\tan^2 x + 1) + 1} dx = \int \frac{\sec^2 x}{\tan^2 x + 2} dx$$

Let $u = \tan x$, meaning $du = \sec^2 x dx$:

$$\int \frac{1}{u^2 + 2} du = \frac{1}{\sqrt{2}} \arctan\left(\frac{u}{\sqrt{2}}\right) + C = \frac{1}{\sqrt{2}} \arctan\left(\frac{\tan x}{\sqrt{2}}\right) + C$$

Example 6

$$\int \sqrt{x} e^{\sqrt{x}} dx$$

Eliminate the radical fraction setting $u = \sqrt{x}$, meaning $u^2 = x$ and $2u du = dx$:

$$\int u \cdot e^u \cdot (2u du) = 2 \int u^2 e^u du$$

Applying tabular integration by parts on $2 \int u^2 e^u du$:

$$2e^u(u^2 - 2u + 2) + C = 2e^{\sqrt{x}}(x - 2\sqrt{x} + 2) + C$$

Example 7: The "Root Elimination" Strategy

$$\int \frac{1}{\sqrt{\sqrt{x} + 1}} dx$$

When dealing with deeply nested radicals, set u to equal the entire expression: $u = \sqrt{\sqrt{x} + 1}$. Squaring both sides yields $u^2 = \sqrt{x} + 1 \implies \sqrt{x} = u^2 - 1$. Squaring again gives $x = (u^2 - 1)^2$. Compute the differential dx :

$$dx = 2(u^2 - 1) \cdot 2u du = 4u(u^2 - 1) du$$

Substitute these back into the integral:

$$\int \frac{1}{u} \cdot 4u(u^2 - 1) du = 4 \int (u^2 - 1) du = 4 \left(\frac{u^3}{3} - u \right) + C$$

Substituting $u = \sqrt{\sqrt{x} + 1}$ back gives:

$$\frac{4}{3}(\sqrt{x} + 1)^{3/2} - 4\sqrt{\sqrt{x} + 1} + C$$

Example 8: Double Angle Chain Rule

$$\int \frac{\sin 2x}{1 + \sin^4 x} dx$$

Recall that the derivative of $\sin^2 x$ is $2 \sin x \cos x = \sin 2x$. Let $u = \sin^2 x$, which gives $du = \sin 2x dx$:

$$\int \frac{1}{1 + u^2} du = \arctan(u) + C = \arctan(\sin^2 x) + C$$

Example 9: Secret Logarithmic Substitution

$$\int \frac{\ln(\tan x)}{\sin x \cos x} dx$$

Test the derivative of the target function. Let $u = \ln(\tan x)$:

$$du = \frac{1}{\tan x} \cdot \sec^2 x dx = \frac{\cos x}{\sin x} \cdot \frac{1}{\cos^2 x} dx = \frac{1}{\sin x \cos x} dx$$

This perfectly eliminates the denominator, turning the problem into a fundamental power-rule integral:

$$\int u du = \frac{1}{2}u^2 + C = \frac{1}{2}(\ln(\tan x))^2 + C$$

8.3 Conjugation and Advanced Definite Integrals

Example 10

$$\int \frac{1}{\sqrt{x+1} + \sqrt{x}} dx$$

Multiply the numerator and denominator by the conjugate expression $\sqrt{x+1} - \sqrt{x}$:

$$\int \frac{\sqrt{x+1} - \sqrt{x}}{(x+1) - x} dx = \int (\sqrt{x+1} - \sqrt{x}) dx$$

Apply the power rule directly:

$$\frac{2}{3}(x+1)^{3/2} - \frac{2}{3}x^{3/2} + C$$

Example 11

$$\int \frac{\ln(x+1)}{x^2} dx$$

Apply integration by parts with $u = \ln(x+1)$ and $dv = \frac{1}{x^2} dx \implies du = \frac{1}{x+1} dx$ and $v = -\frac{1}{x}$:

$$-\frac{\ln(x+1)}{x} + \int \frac{1}{x(x+1)} dx$$

Using a zero-substitution $(+x - x)$ inside the remaining integral yields $\int \left(\frac{1}{x} - \frac{1}{x+1}\right) dx$:

$$-\frac{\ln(x+1)}{x} + \ln|x| - \ln|x+1| + C$$

Example 12

$$\int 2^x \cdot 5^x dx$$

Consolidate the base terms utilizing standard exponent algebra rules:

$$\int (2 \cdot 5)^x dx = \int 10^x dx = \frac{10^x}{\ln 10} + C$$

Example 13

$$\int \frac{1}{x \ln x - x} dx$$

Factor out x from the denominator to establish a clear u -substitution pathway:

$$\int \frac{1}{x(\ln x - 1)} dx$$

Let $u = \ln x - 1 \implies du = \frac{1}{x} dx$:

$$\int \frac{1}{u} du = \ln|u| + C = \ln|\ln x - 1| + C$$

Example 14

$$\int \sqrt{1 - \sin x} \, dx$$

Multiply the expression inside the radical by the trigonometric conjugate $\frac{1+\sin x}{1+\sin x}$:

$$\int \sqrt{\frac{1 - \sin^2 x}{1 + \sin x}} \, dx = \int \frac{\cos x}{\sqrt{1 + \sin x}} \, dx$$

Let $u = 1 + \sin x \implies du = \cos x \, dx$:

$$\int \frac{1}{\sqrt{u}} \, du = 2\sqrt{u} + C = 2\sqrt{1 + \sin x} + C$$

Example 15

$$\int \frac{x^2 + 2}{x + 2} \, dx$$

Perform a nested zero-substitution to perfectly divide the polynomial without long division. Add and subtract $2x$ then add and subtract 4:

$$\int \frac{x^2 + 2x - 2x - 4 + 6}{x + 2} \, dx = \int \frac{x(x + 2) - 2(x + 2) + 6}{x + 2} \, dx = \int \left(x - 2 + \frac{6}{x + 2} \right) \, dx$$

Integrating the simplified polynomial terms:

$$\frac{x^2}{2} - 2x + 6 \ln |x + 2| + C$$

Example 16: Rationalizing Fractional Powers

$$\int_{-1}^0 \frac{\sqrt[3]{x} + 1}{\sqrt[3]{x} - 1} \, dx$$

Let $u = \sqrt[3]{x} \implies u^3 = x \implies dx = 3u^2 \, du$. Update the bounds: $x = -1 \implies u = -1$ and $x = 0 \implies u = 0$:

$$\int_{-1}^0 \frac{u + 1}{u - 1} \cdot 3u^2 \, du = 3 \int_{-1}^0 \frac{u^3 + u^2}{u - 1} \, du$$

Using polynomial division, rewrite the integrand:

$$3 \int_{-1}^0 \left(u^2 + 2u + 2 + \frac{2}{u - 1} \right) \, du = 3 \left[\frac{u^3}{3} + u^2 + 2u + 2 \ln |u - 1| \right]_{-1}^0$$

Evaluating at the upper and lower limits results in the final scalar value:

$$4 - 6 \ln 2$$

Example 17: Symmetry Trap with Absolute Value

$$\int_{-3}^3 \frac{x|x|}{x^2 + 1} \, dx$$

Before performing any algebraic steps, analyze the bounds $[-3, 3]$ for symmetry. The absolute value function $|x|$ is even, and $x^2 + 1$ is even, making their combination even. However, the

isolated x in the numerator is an odd function. An odd function multiplied by an even function creates an overall odd integrand:

$$f(-x) = \frac{-x|x| - x|}{(-x)^2 + 1} = -\frac{x|x|}{x^2 + 1} = -f(x)$$

Because the integrand is an odd function integrated over symmetric bounds, it evaluates directly to zero:

$$\int_{-3}^3 \frac{x|x|}{x^2 + 1} dx = 0$$

Example 18: Advanced Rational Exponential Substitution

$$\int_0^{\ln 10} \frac{e^x \sqrt{e^x - 1}}{e^x + 8} dx$$

Let $u = \sqrt{e^x - 1} \implies u^2 = e^x - 1 \implies 2u du = e^x dx$. Note that $e^x = u^2 + 1$, so the denominator becomes $(u^2 + 1) + 8 = u^2 + 9$. Convert the boundary conditions: $x = 0 \implies u = 0$ and $x = \ln 10 \implies u = \sqrt{10 - 1} = 3$:

$$\int_0^3 \frac{u \cdot (2u du)}{u^2 + 9} = 2 \int_0^3 \frac{u^2}{u^2 + 9} du$$

Apply a zero-substitution $(+9 - 9)$ inside the numerator:

$$2 \int_0^3 \left(1 - \frac{9}{u^2 + 9} \right) du = 2 \left[u - 3 \arctan \left(\frac{u}{3} \right) \right]_0^3$$

Substitute the boundaries:

$$2(3 - 3 \arctan(1)) - 0 = 2 \left(3 - \frac{3\pi}{4} \right) = 6 - \frac{3\pi}{2}$$

Example 19: Radical Fractional Grouping

$$\int \frac{1}{\sqrt{x + x^{3/2}}} dx$$

Factor out x inside the square root to parse the functions cleanly:

$$\int \frac{1}{\sqrt{x(1 + \sqrt{x})}} dx = \int \frac{1}{\sqrt{x} \sqrt{1 + \sqrt{x}}} dx$$

Let $u = 1 + \sqrt{x} \implies du = \frac{1}{2\sqrt{x}} dx \implies 2 du = \frac{1}{\sqrt{x}} dx$:

$$\int \frac{2}{\sqrt{u}} du = 4\sqrt{u} + C = 4\sqrt{1 + \sqrt{x}} + C$$

Example 20: Tangent Fraction Component Splitting

$$\int \frac{1 - \tan x}{1 + \tan x} dx$$

Convert the tangents into fundamental sine and cosine functions:

$$\int \frac{1 - \frac{\sin x}{\cos x}}{1 + \frac{\sin x}{\cos x}} dx = \int \frac{\cos x - \sin x}{\cos x + \sin x} dx$$

Let $u = \cos x + \sin x$. The differential is $du = (-\sin x + \cos x) dx$, which perfectly matches the numerator:

$$\int \frac{1}{u} du = \ln |u| + C = \ln |\cos x + \sin x| + C$$